

- HCL Digital Experience – OpenShift Installation Guide
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HCL Digital Experience – OpenShift Installation Guide

DX 9.5 CF231 – EuropeSIP Lab Cluster

Before starting the installation, the OpenShift administrator must ensure that the environment meets all required technical prerequisites. This involves preparing the necessary storage, configuring the container registry, setting up DNS entries, and providing an appropriate workstation from which the installation will be performed.

The administrator must complete the following tasks **before** beginning the installation:

✓ A Linux-based workstation / bastion

A workstation is required to run all installation commands. Any of the following options are valid:

- Linux workstation
- WSL2 on Windows
- A bastion VM with `kubectl`, `oc`, and `helm` installed

This workstation must include the following tools:

Required tools:

- `openssl` — needed to generate TLS certificates
- `git` — required if cloning the lab repository

Optional but recommended tools:

- `podman` or `docker` — useful for inspecting or manipulating container images
- `skopeo` — for copying, inspecting, or syncing images between registries
- `curl` or `wget` — for fetching files, endpoints, and testing connectivity
- `stern` or `kubetail` — for aggregated log streaming across multiple Kubernetes pods
- `jq` — for parsing and inspecting JSON output from Kubernetes/oc commands

These optional tools are not required for the installation but are highly useful for troubleshooting, debugging, and working efficiently in Kubernetes/OpenShift environments.

In addition, the workstation must have:

- Network access to the OpenShift cluster (which may operate in an air-gapped mode if required)
- Optional, but not strictly necessary, access to the Internet
- Access to the image registry being used
- Access to the DX Helm chart package (`hcl-dx-deployment-2.42.1.tgz`) locally or via shared storage

✓ Required StorageClasses

The cluster must provide at least one **RWX** StorageClass and one **RWO** StorageClass for the DX deployment.

In this guide, we will use:

- `dx-deploy-rwx` — shared (ReadWriteMany) volumes
- `dx-deploy-rwo` — persistent (ReadWriteOnce) volumes

Note: You may use any existing StorageClass names in your environment by updating them in `custom-values.yaml`. Since StorageClass provisioning differs across environments (NFS, ODF, SAN, cloud storage, etc.), **their creation and configuration are outside the scope of this guide** and are the responsibility of the OpenShift administrator.

✓ Container images available in the registry

The OpenShift administrator must preload all required HCL DX container images into the container registry used by the environment.

This guide assumes a restrictive scenario in which the OpenShift cluster does **not** have Internet access (air-gapped environment). Therefore, the cluster cannot pull images directly from the HCL Harbor registry. Instead, all images must be retrieved from a local corporate registry where they have been previously mirrored or uploaded by the administrator.

As an example of the process to upload the images to your local corporate registry, you must follow the instructions [here](#)

This process may involve:

- Pulling DX images from HCL's distribution source
- Tagging and pushing them into the organization's registry
- Ensuring OpenShift worker nodes can pull the images
- Creating ImagePullSecrets if authentication is required

Important: Because registry configuration varies significantly between customers and deployments — and because corporate environments typically maintain their own centralized registries — **the process of uploading images to the registry is not covered in this guide**. It is assumed that all required DX images are already present and accessible to the cluster.

✓ DNS hostname prepared

Before installing DX, the DNS administrator must configure the hostname that will be used to access the DX environment.

This includes:

- Creating the appropriate DNS record (typically an A or CNAME record)
- Pointing it to the OpenShift ingress endpoint or load balancer
- Ensuring that the hostname resolves correctly from the installer workstation and user networks

Note: DNS configuration procedures vary across environments (internal DNS, cloud DNS, external providers, etc.), therefore **the steps to create DNS entries are outside the scope of this guide.**

✓ TLS certificates (self-signed or corporate)

This guide uses self-signed TLS certificates generated during the installation process. These certificates are suitable for lab environments and internal testing scenarios.

However, if you prefer to use corporate or pre-existing certificates issued by your organization or a trusted Certificate Authority, the OpenShift administrator must provide the corresponding *.pem files **before** beginning the installation.

These files should include:

- The certificate (.pem or .crt)
- The private key (.key or .pem)
- Any required intermediate or CA certificates, when applicable

During the installation, the dxadmin user can load these certificates into OpenShift as a TLS secret instead of generating self-signed ones.

Note: The process for obtaining or approving corporate certificates varies by organization and is therefore outside the scope of this guide.

✓ Installer user (dxadmin)

This guide assumes that the dxadmin user already exists with the appropriate permissions to deploy DX inside the designated namespace.

Because user creation depends on the identity provider configured in the cluster (Azure AD, Keycloak, LDAP, htpasswd, etc.), **the creation and configuration of the dxadmin account are outside the scope of this guide** and must be performed by the OpenShift administrator before starting the installation.

✓ Optional: Clone the lab repository (recommended)

To simplify the installation process, you may optionally clone the public GitHub repository that contains:

- Updated installation instructions
- Example configuration files
- Sample `custom-values-sample.yaml`
- Route definitions
- Utility scripts used throughout the lab
- Troubleshooting helpers

Clone the repository with:

```
git clone https://github.com/europesip/dx-deployment.git
cd dx-deployment
```

After cloning the repository, you may also copy or download the Helm chart required for the installation (`hcl-dx-deployment-2.42.1.tgz`, as referenced at the beginning of this document). The Helm deployment package is available through HCL Software Downloads at: <https://my.hcltechsw.com/downloads>